

When Good Probes Fail

Stress-Testing an Efficient Hidden-State Robustness Predictor

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Abstract

We test whether inexpensive hidden-state probes can predict if a mathematical reasoning model remains correct under meaning-preserving perturbations. Our V6 submission applies balanced linear probes to nine fixed layers of DeepSeek-R1-0528-Qwen3-8B and aggregates 225 hard votes. It achieved 0.705 grouped out-of-fold (OOF) balanced accuracy and 12/19 Small Track private accuracy with a 4.23 MB bundle. Four pre-registered variants produced only small or unstable in-distribution gains. A frozen audit on ten non-overlapping AIMO problems then found that V6 predicted every problem robust, scoring 0.20 accuracy against an 0.80 always-negative baseline. Thus representation separability and a small private-set gain did not establish OOD reliability. We contribute an auditable low-resource workflow, controlled negative results, and safeguards for interpretability research under small, imbalanced datasets.

1. Motivation

The AIMO Interpretability Challenge asks whether a system can distinguish robust mathematical reasoning from solutions that fail under controlled counterfactual variation [1]. Probing frozen internal representations is compute-efficient, but probe accuracy is correlational and may reflect dataset regularities or probe capacity rather than a mechanism used by the model [3,4]. We therefore ask whether apparent probe advantage survives changes in problem source and evaluation distribution.

2. Data audit

The official aggregate data had 141 rows but only 137 unique problem IDs; four were exact, label-consistent duplicates. After collapse, labels were 101 non-robust and 36 robust. The official 675-row expanded data still contained the same 137 problems, so perturbation rows were treated as repeated measurements rather than independent samples. Every split was grouped by problem ID, and preprocessing was fit inside training partitions.

The published binary label was reproduced on all 137 problems by marking a problem robust only when its maximum absolute accuracy decay across available perturbations was at most zero. Revisions and canonical content hashes were frozen before evaluation.

3. Frozen V6 method

For each problem, V6 extracts the final prompt-token state from the frozen 8B checkpoint at layers 4, 8, ..., 36. Each layer receives a balanced L2 logistic probe with $C=0.001$ after training-only standardization. Five grouped folds under five fixed seeds yield 25 fold groups $\times$ 9 layers = 225 deployment votes. The final threshold is a fixed majority vote. The method uses one model forward pass, no generation or external API during scoring, and a 4.23 MB submission bundle.

4. Validation protocol

V6 used repeated grouped OOF validation, both directional source holdouts, a shuffled-label control, 5,000 problem bootstrap samples, and exact deployment replay. Because prior exploratory results had been observed, V6 public evidence is sequentially exploratory, not untouched confirmation.

E6-E9 each changed one component and were registered before evaluation. Promotion required at least +0.02 balanced accuracy over V6, a positive paired bootstrap lower bound, strong performance in each source direction, advantage over shuffled labels, and nontrivial prediction disagreement.

5. Public and private results

Exp.	Change from V6	OOF BA	Decision
V6	Final-token linear vote	0.7048	Exploratory pass
E6	Adjacent-layer deltas	0.7137	Reject
E7	Mean problem-token state	0.7186	Reject
E8	Final problem-token state	0.6949	Reject
E9	PCA10 + RBF vote	0.7126	Reject

V6's public confusion was $TN=61$, $FP=40$, $FN=7$, $TP=29$. Its one-sided bootstrap 95% lower bound was 0.6169 and its advantage over the shuffled-label control was 0.1189. On 19 private Small Track cases it scored 12/19 with full coverage and no invalid predictions - the first non-constant private improvement in our sequence. The sample is too small to establish reliable generalization.

No successor passed. E6 changed only 2/137 decisions. E7 had the highest OOF balanced accuracy but asymmetric source transfer (0.612 vs 0.756). E8 weakened aggregate and transfer evidence. E9 raised ordinary accuracy to 0.708, yet its paired balanced-accuracy interval crossed zero and one source direction fell to 0.628. Accuracy and balanced accuracy must both be reported under the 101:36 imbalance.

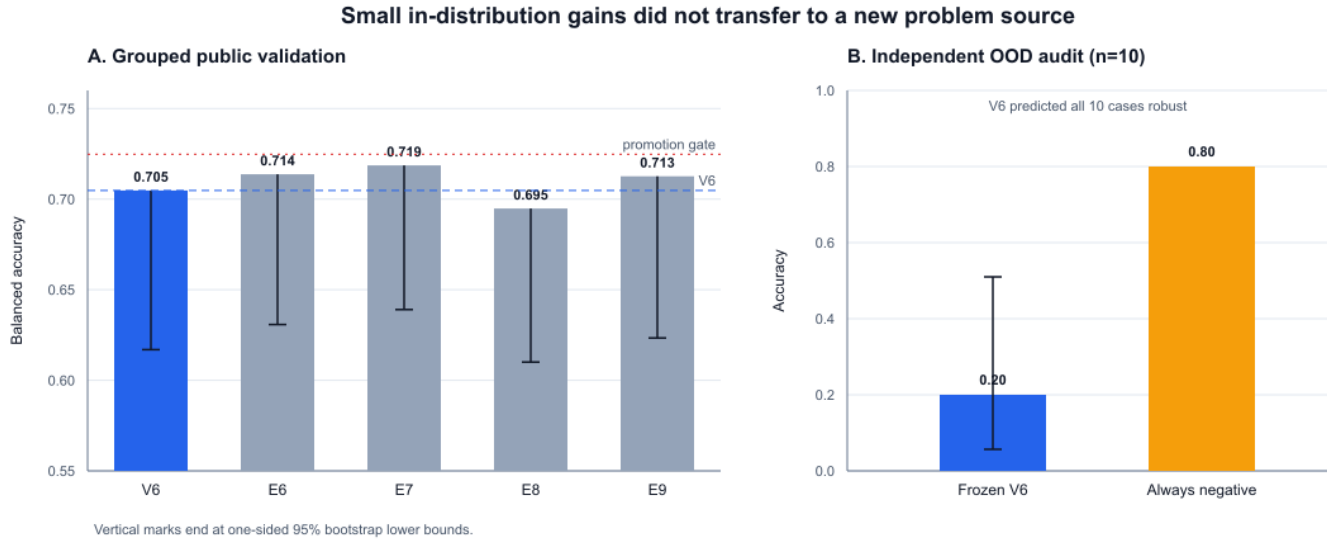


Figure 1. Public grouped validation gives several small apparent gains, but all registered successors fail promotion and frozen V6 collapses on the independent ten-problem OOD audit. Error bar: 95% Wilson interval.

6. Independent OOD audit

E10 evaluated the already frozen V6 artifact on the direct qwen3-8b:low slice of a separately published ten-problem AIMO sample [2]. Its 64 perturbation rows yielded eight non-robust and two robust problem labels. The label rule reproduced all 137 training labels, and overlap was zero by ID and normalized text. Labels were never supplied to the model or probe.

Before extraction, we pinned dataset, model, and artifact hashes and required at least eight two-class problems, accuracy ≥ 0.60 , and $\geq +0.10$ over the best constant predictor. V6 predicted all ten cases robust. Accuracy was 0.20, balanced accuracy 0.50, and the 95% Wilson interval [0.057, 0.510]; always-negative accuracy was 0.80. Median robust-vote fraction was 0.94 and the minimum 0.769, indicating confident OOD error rather than a small threshold mismatch.

The audit failed by its registered rule. We did not tune on individual E10 cases and did not create V8.

7. Actionable lessons

1. Split at the causal unit: variants of one problem are repeated measurements.
2. Random-label controls are necessary but insufficient.
3. Small gains need paired evidence.
4. Source-direction symmetry is an early warning.
5. Confidence under shift distinguishes a threshold near-miss from extrapolation failure.
6. Pre-registered stopping rules turn tempting marginal gains into useful negative results.

8. Limitations and claim boundary

Training used 137 problems from two related MATH sources; private and OOD samples contained only 19 and ten cases. E10 tests problem transfer for one 8B checkpoint, not all models or perturbations. Linear separability does not show that the model uses the probed features causally. Our defensible claim is negative: final-token states contain in-distribution robustness signal, but this probe family is not supported as a general detector of robust mathematical reasoning.

9. Reproducibility and resources

Inputs, registrations, artifacts, and result JSONs are hash-pinned. Activation extraction used free Kaggle T4 x 2 sessions; probe training, bootstrap, packaging, and audit ran locally on CPU, incurring zero direct monetary compute cost. Any future model must start from a distinct mechanistic hypothesis and an untouched validation source; V6 private outcomes and E10 are frozen against tuning.

10. Conclusion

An efficient hidden-state probe showed meaningful public and small-private-set signal, but plausible refinements failed strict paired gates and the frozen method collapsed on a new AIMO problem source. The gap between separability and transfer is the main result.

References

- [1] Stefanik et al. *AIMO Interpretability Challenge*. arXiv:2607.13899, 2026. <https://arxiv.org/abs/2607.13899>
- [2] AIMO challenge sample-full dataset. <https://huggingface.co/datasets/aimo-interp/aimo-interp-challenge-sample-full>
- [3] Belinkov. *Probing Classifiers: Promises, Shortcomings, and Advances*. Computational Linguistics, 2022. <https://aclanthology.org/2022.cl-1.7/>
- [4] Hewitt and Liang. *Designing and Interpreting Probes with Control Tasks*. EMNLP-IJCNLP, 2019. <https://aclanthology.org/D19-1275/>